

Microbial Interaction Networks: Implications for Ecosystem Health and Emerging Biological Risks

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Abstract

This study examined the microbiome of female *Hyalomma excavatum* ticks from cattle in Djelfa across three seasons using 16S rRNA sequencing. A total of 640 bacterial taxa were identified, showing clear seasonal variation. The results reveal dynamic, environmentally driven microbial shifts with potential applications in biotechnology and disease surveillance.

Introduction

Ticks harbor diverse microbial communities that influence their physiology and ability to transmit pathogens [1].

Hyalomma excavatum, a widespread tick species in Algeria, remains poorly characterized in terms of its associated microflora and seasonal variability [2].

Objective: To characterize microbial diversity and interaction patterns in female *Hyalomma excavatum* from the Algerian steppe and assess the potential relevance of microbial interaction networks for ecosystem health and biological risk surveillance.

Material and Methods

Sampling: Female *H. excavatum* collected from cattle in Djelfa (central Algerian steppe) during autumn, spring, and summer.

Processing: Surface sterilization and individual DNA extraction.

Sequencing: 16S rRNA gene (V3–V4 region) using high-throughput sequencing.

Analysis:

Taxonomic profiling of bacterial communities.

Alpha and beta diversity analyses to assess seasonal variation.

Network construction to identify co-occurrence and co-exclusion patterns.

16S rRNA Gene Sequencing

Bioinformatic Processing



Illumina MiSeq platform

Quality filtering

ASV clustering

Taxonomic assignment

Results and discussion

Diversity: A total of 640 bacterial taxa were identified across all samples.

Dominant taxa: The microbiome was dominated by Francisella-like endosymbionts (FLE), forming the core bacterial community.

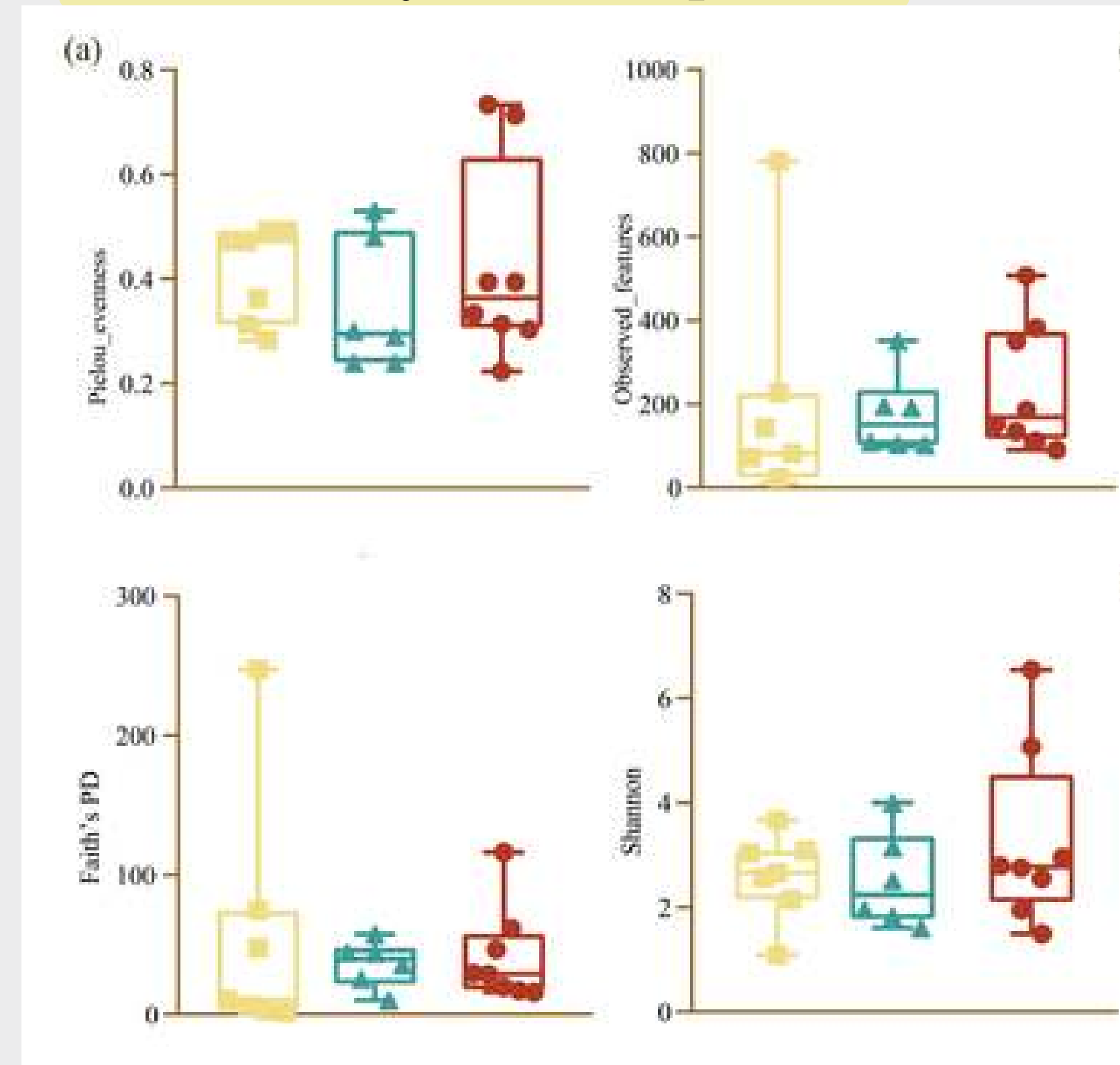
Seasonal variation:

Spring and summer: Higher richness and cooperative associations.

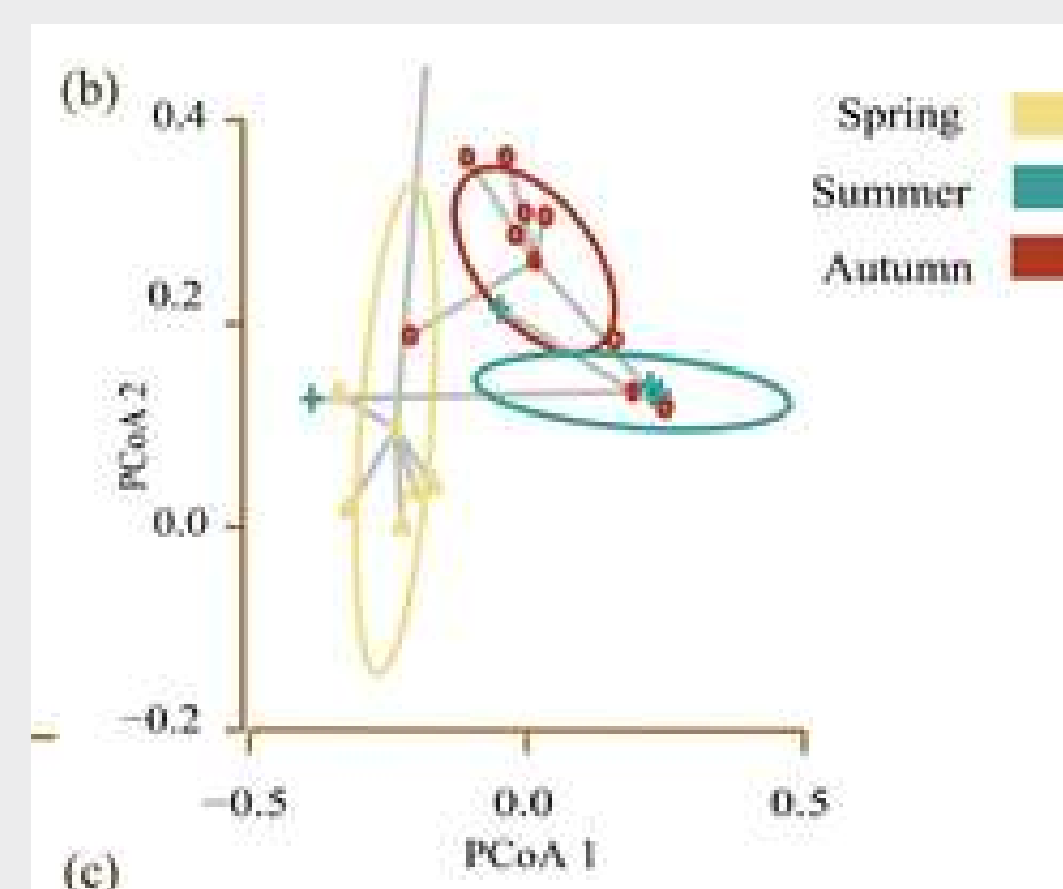
Autumn: Reduced diversity with more competitive interactions.

Network analysis: Microbial interaction structures varied by season, reflecting dynamic community organization influenced by environmental factors.

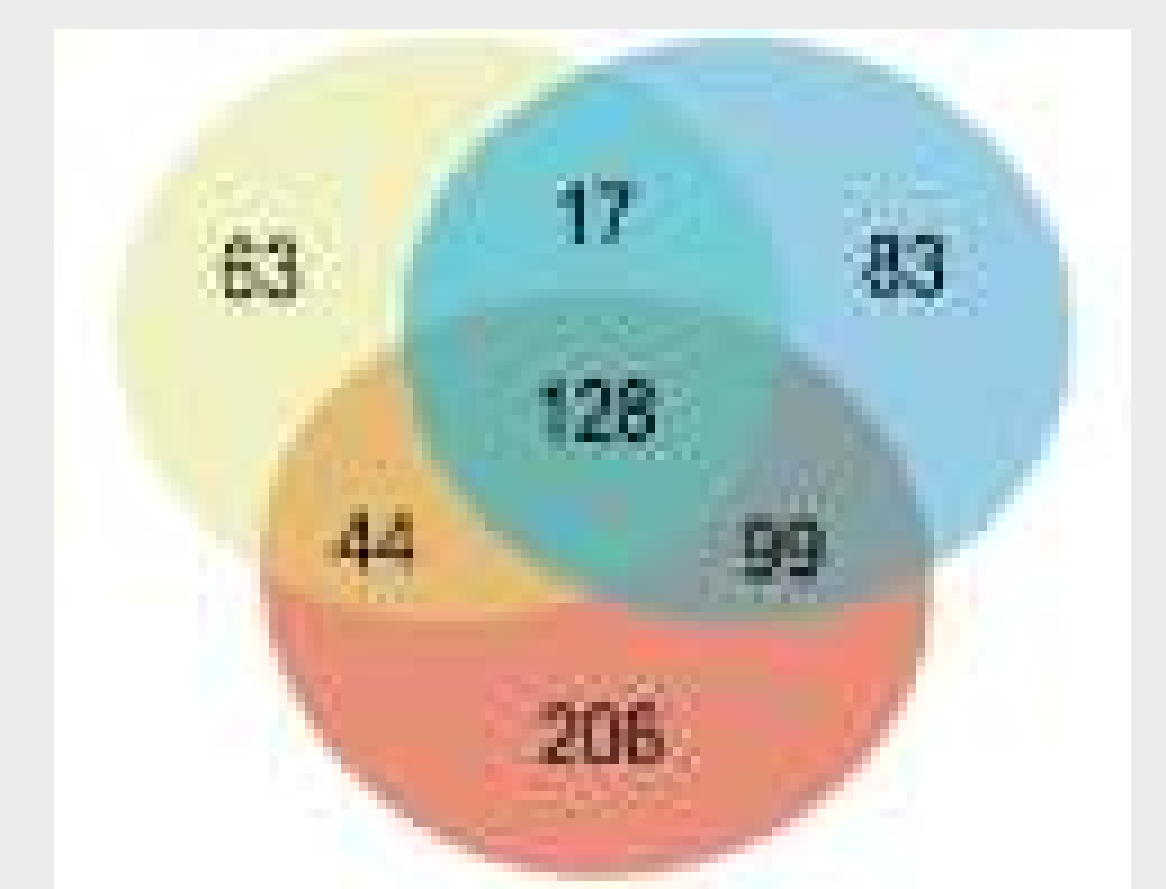
Diversity and Composition



Beta diversity



Alpha diversity



Venn Diagram

Co-occurrence Network Construction



Conclusion

Microbial network analysis revealed positive and negative associations among microorganisms associated with *Hyalomma excavatum*. Seasonal variation influenced network organization, highlighting the dynamic nature of these microbial communities. Network-based surveillance may complement pathogen detection and contribute to biological risk assessment and One Health strategies.

References

1. Perveen N. et al. (2022) Microorganisms 10(11): 2220.
2. Mechouk N. et al. (2022) Parasites Vectors 15:302.